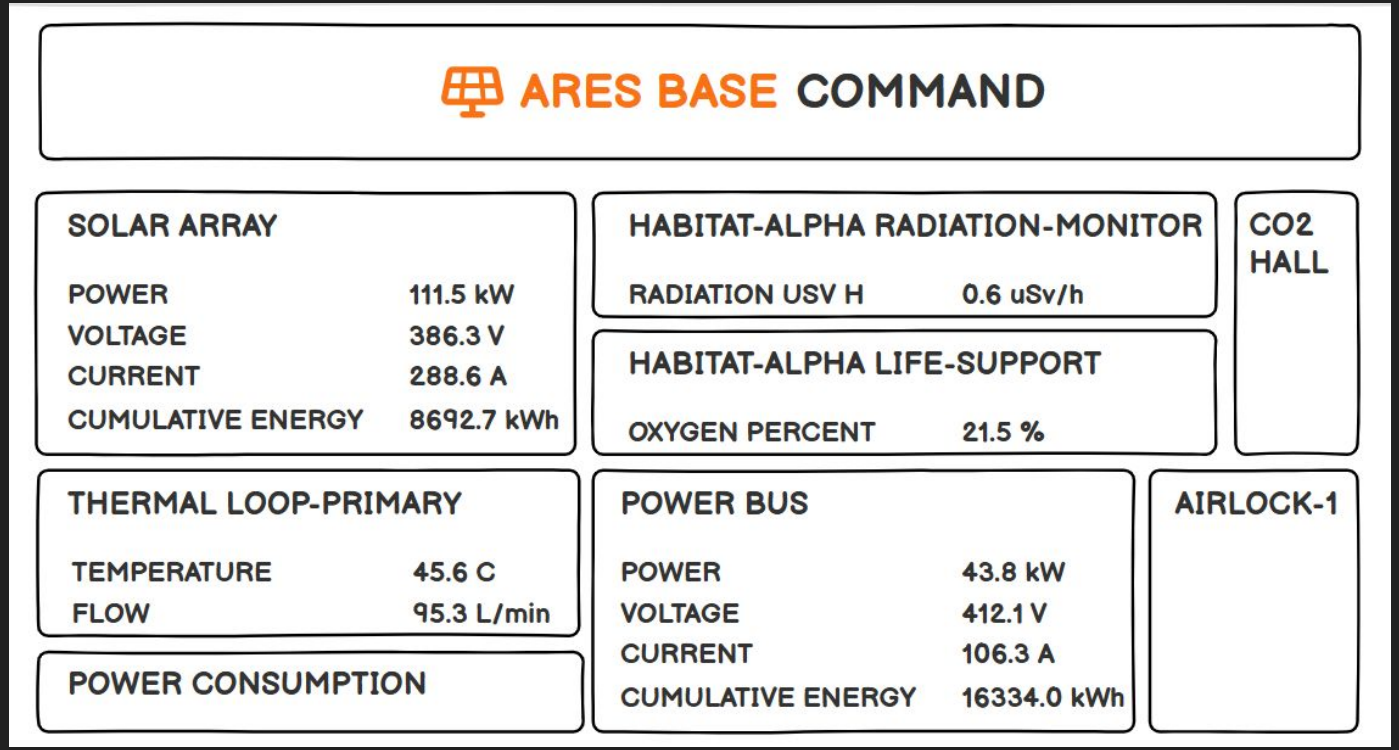
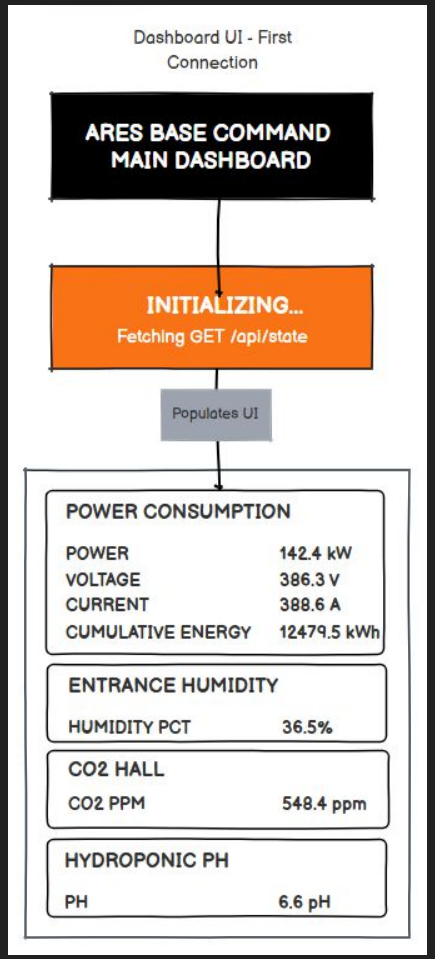
As a Habitat Operator, I want to access a unified main dashboard (`/all`) so that I can see both sensors and telemetry streams together.



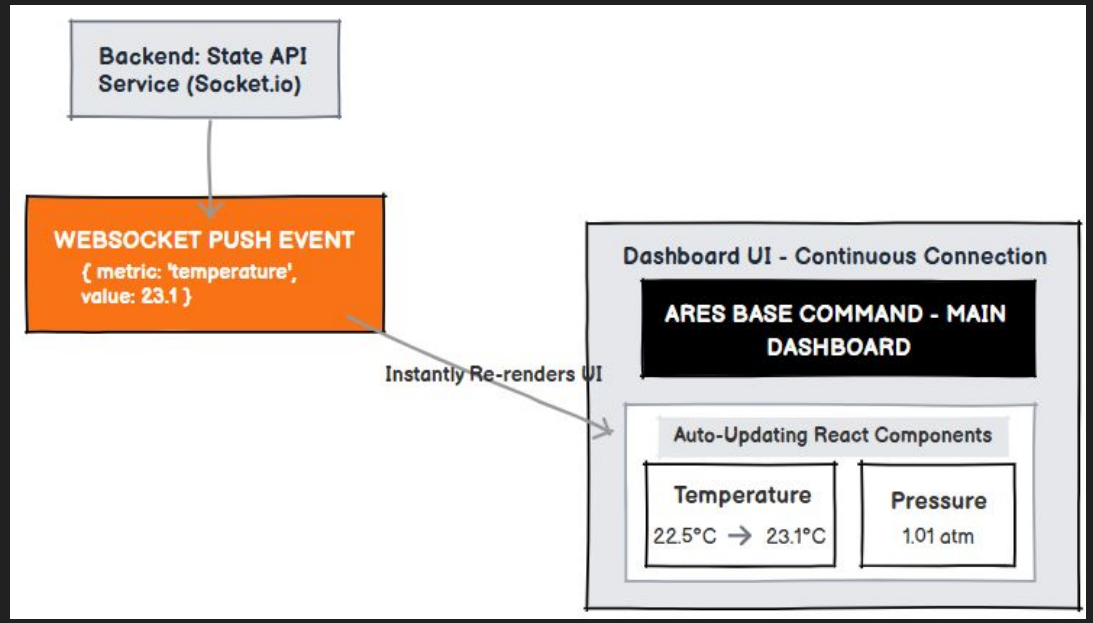
As a Habitat Operator, I want a dedicated "Sensors" page so that I can filter the view and focus only on internal habitat REST metrics.

As a Habitat Operator, I want a dedicated "Telemetry" page so that I can filter the view and focus only on external streams.

As a Habitat Operator, I want the dashboard to fetch the initial state upon first connection (via `GET /api/state`) so that I immediately see the current habitat situation without waiting for new events.

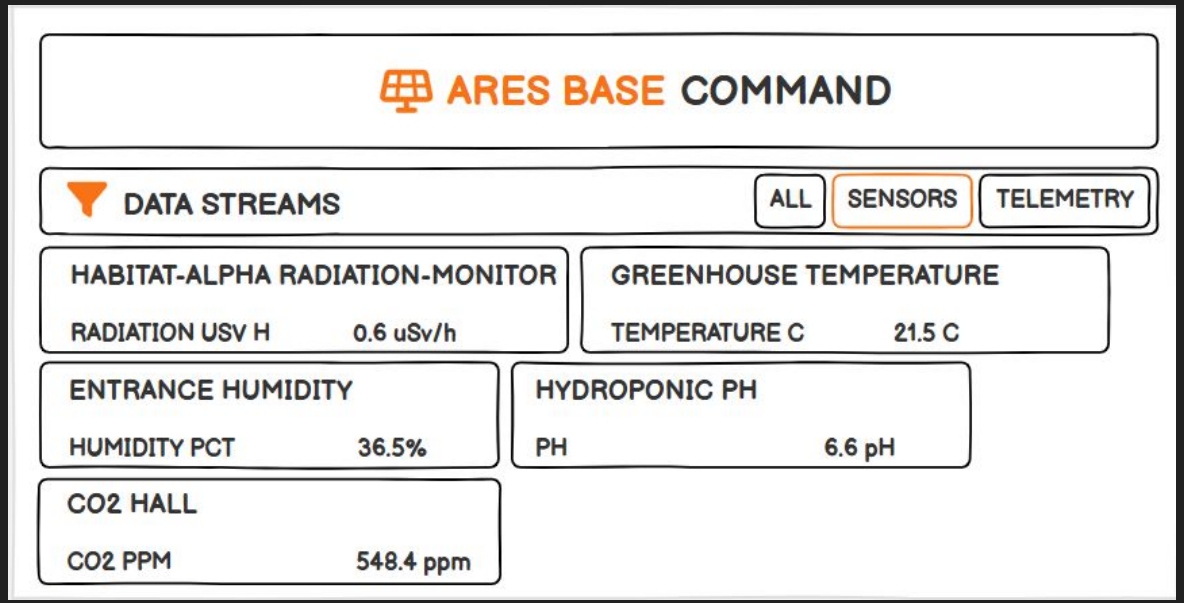


As a Habitat Operator, I want the dashboard to receive data updates via WebSocket (Socket.io) so that I do not have to manually refresh the web page to see new values. **(NFR: Data visualization must rely on full-duplex WebSocket connections instead of HTTP polling)**

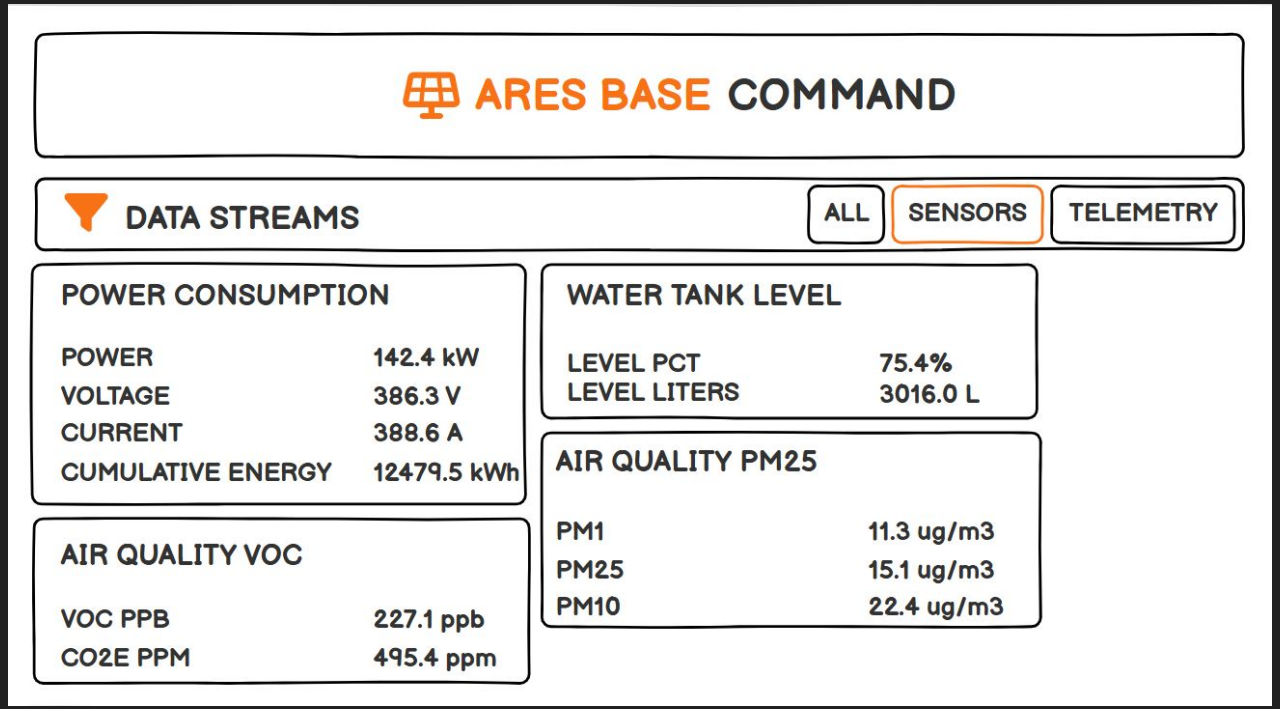


As a Habitat Operator, I want to check the connection with the server from the dashboard via a health status, so that I know if I am really connected.

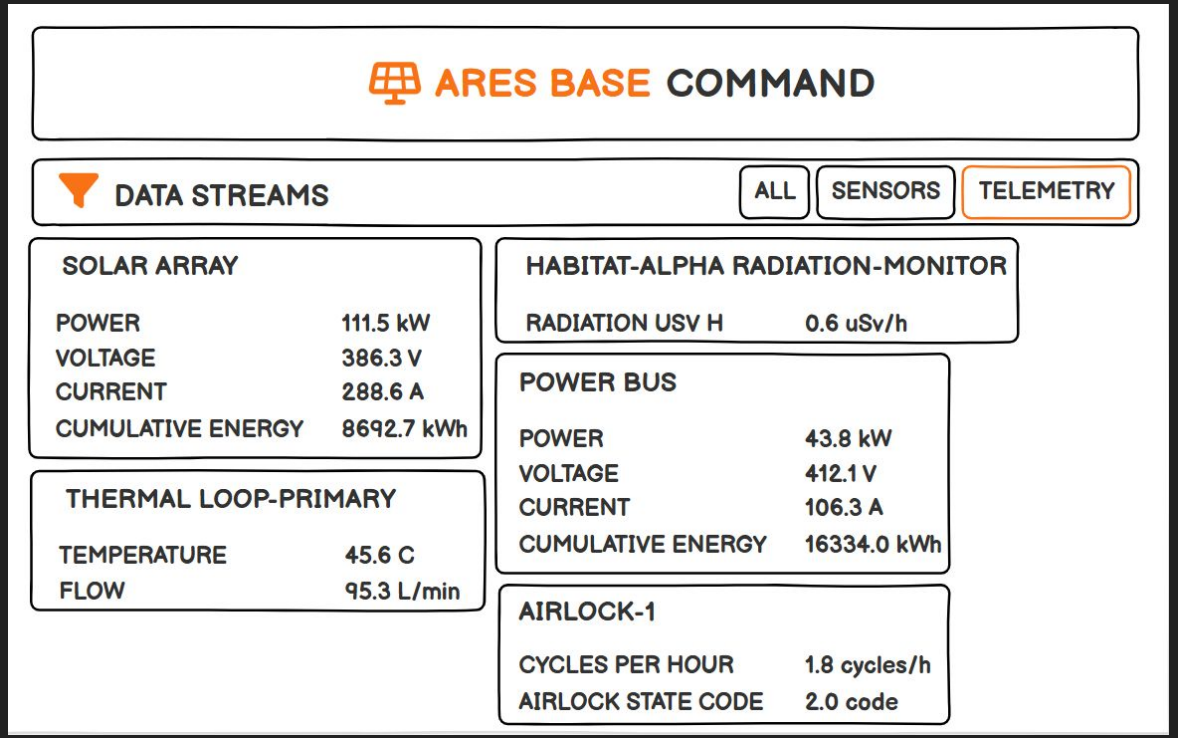
As a Habitat Operator, I want to monitor scalar REST sensors in real-time so that vital internal parameters are maintained.



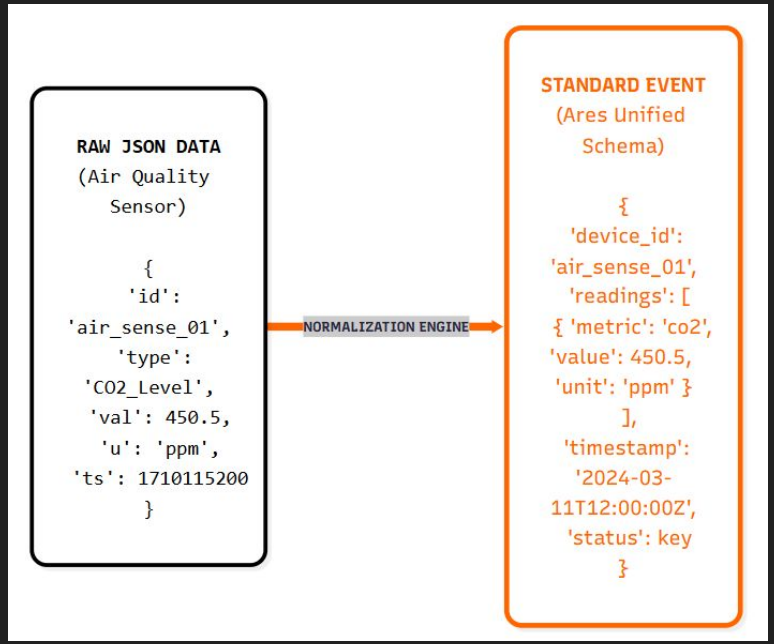
As a Habitat Operator, I want to view complex multi-metric sensors in unified widgets so that I can easily read grouped data.



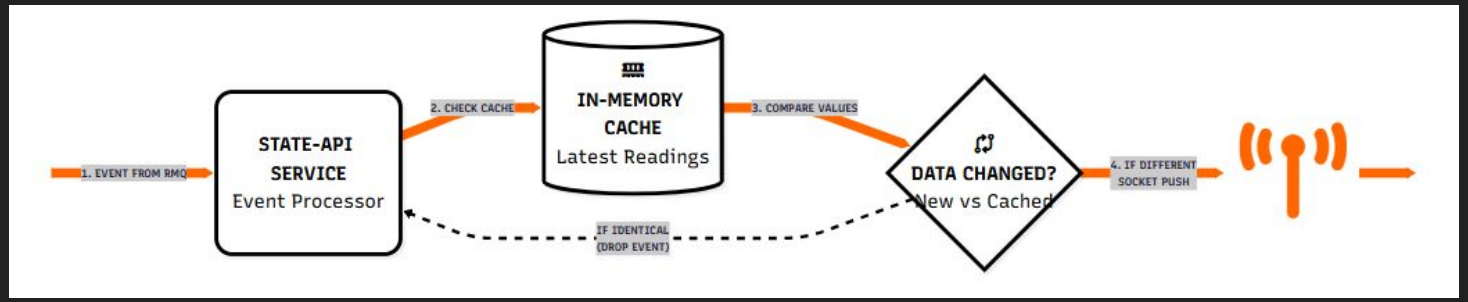
As a Habitat Operator, I want to monitor high-frequency telemetry streams so that I can react instantly to external anomalies.



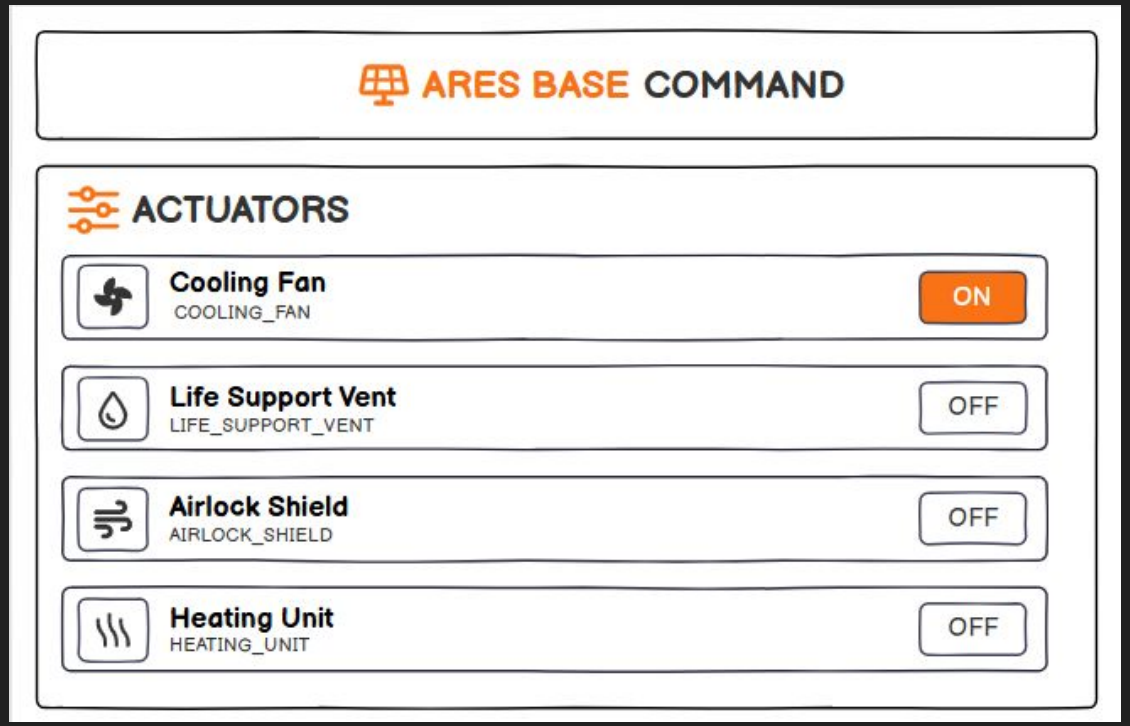
As a System Administrator, I want the backend to normalize all heterogeneous REST and Stream payloads into a standard internal event format so that the system processes everything uniformly. **(NFR: Strict architectural adherence to the unified internal StandardEvent schema).**



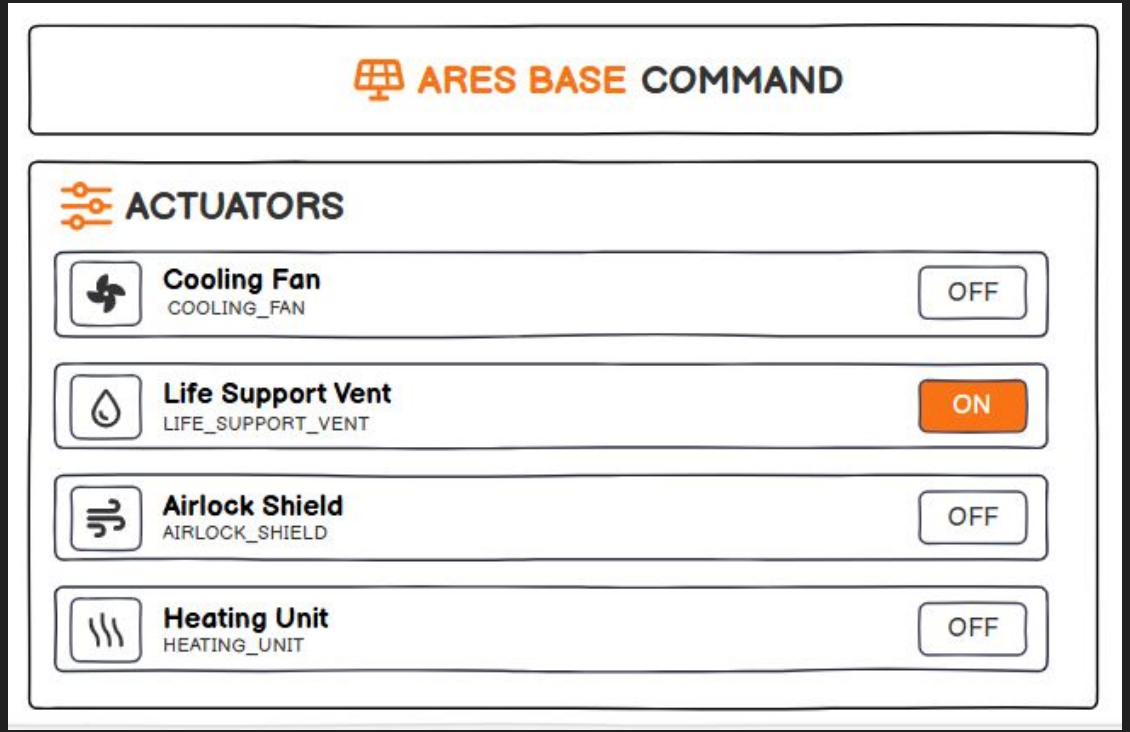
As a System Administrator, I want the State API service to maintain an in-memory cache of the latest readings so that the amount of network calls can be reduced.



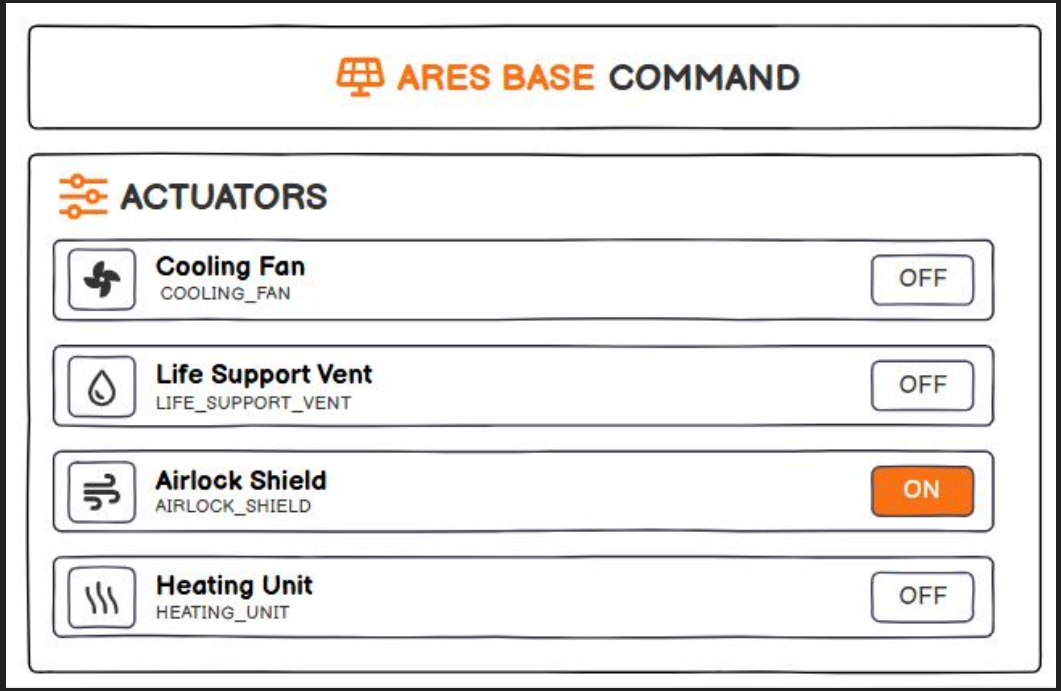
As a Habitat Operator, I want a dashboard button to manually toggle the cooling_fan so that I can force a temperature drop.



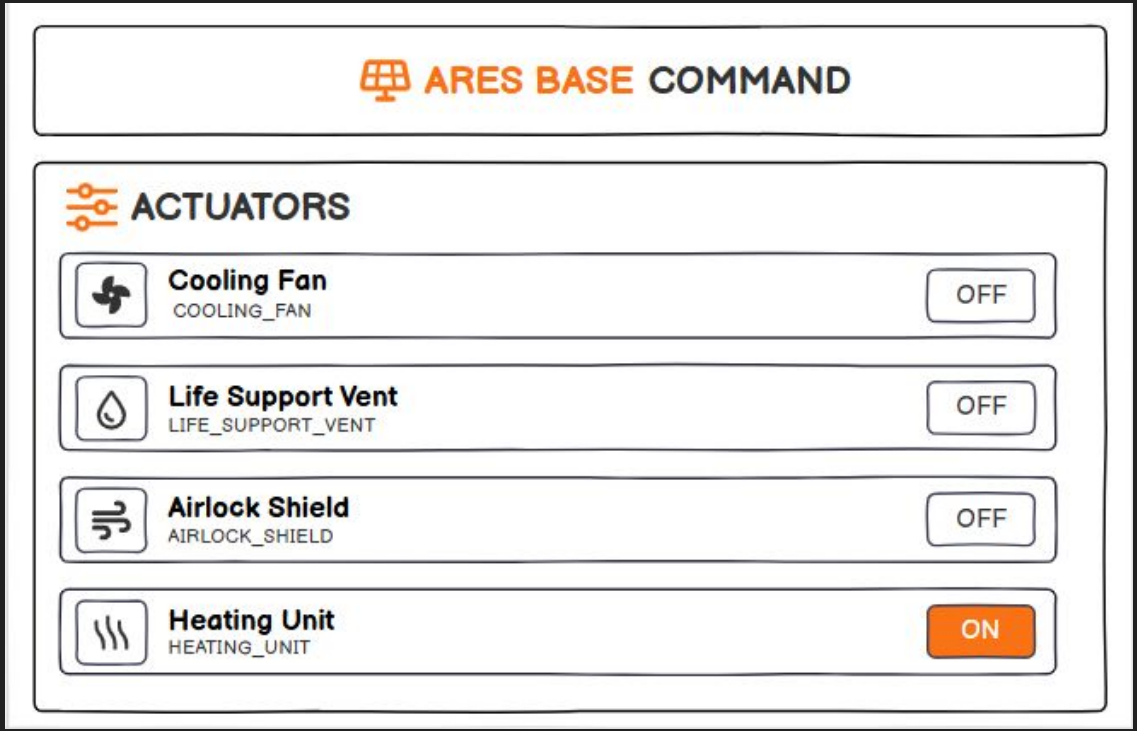
As a Habitat Operator, I want to be able to manually toggle the life_support_vent so that I can correct oxygen levels.



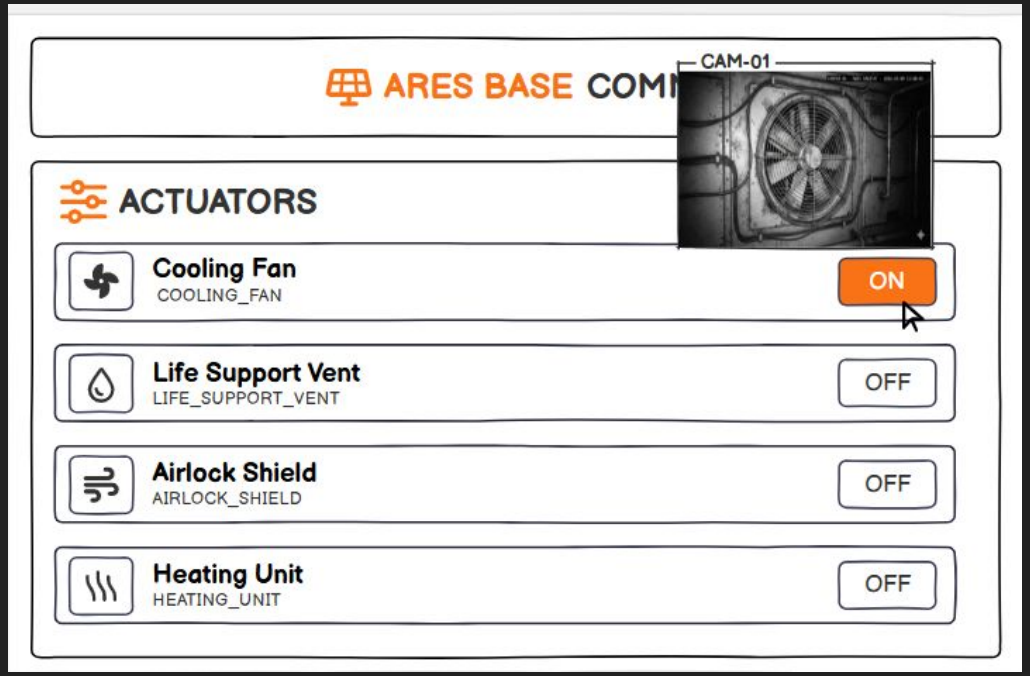
As a Habitat Operator, I want to be able to manually toggle the airlock_shield so that I can protect the base from radiation.



As a Habitat Operator, I want to manually toggle the heating_unit so that the temperature does not drop below the survival threshold.

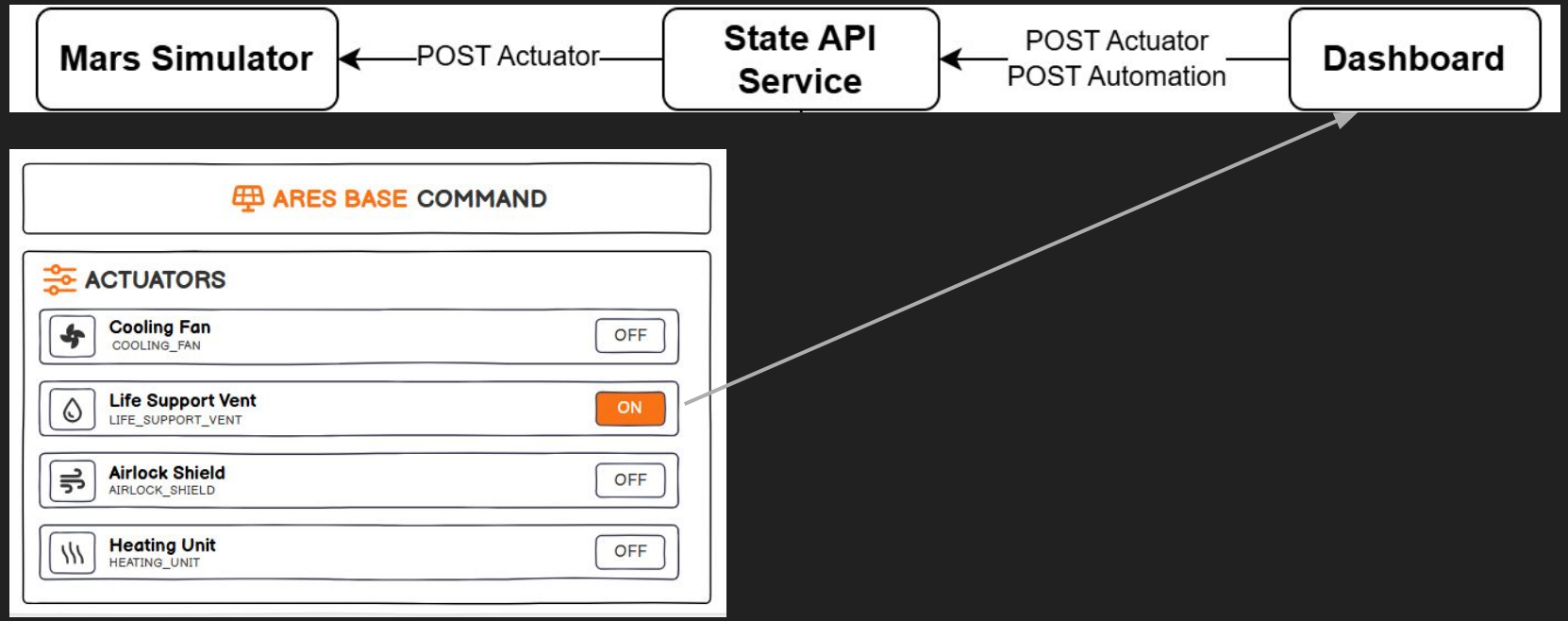


As a Habitat Operator, I want to visually monitor the actuators through a real-time surveillance video-camera, so that I know everything is fine. (**NFR: Video feed components must not block or degrade the performance of real-time telemetry UI rendering)**

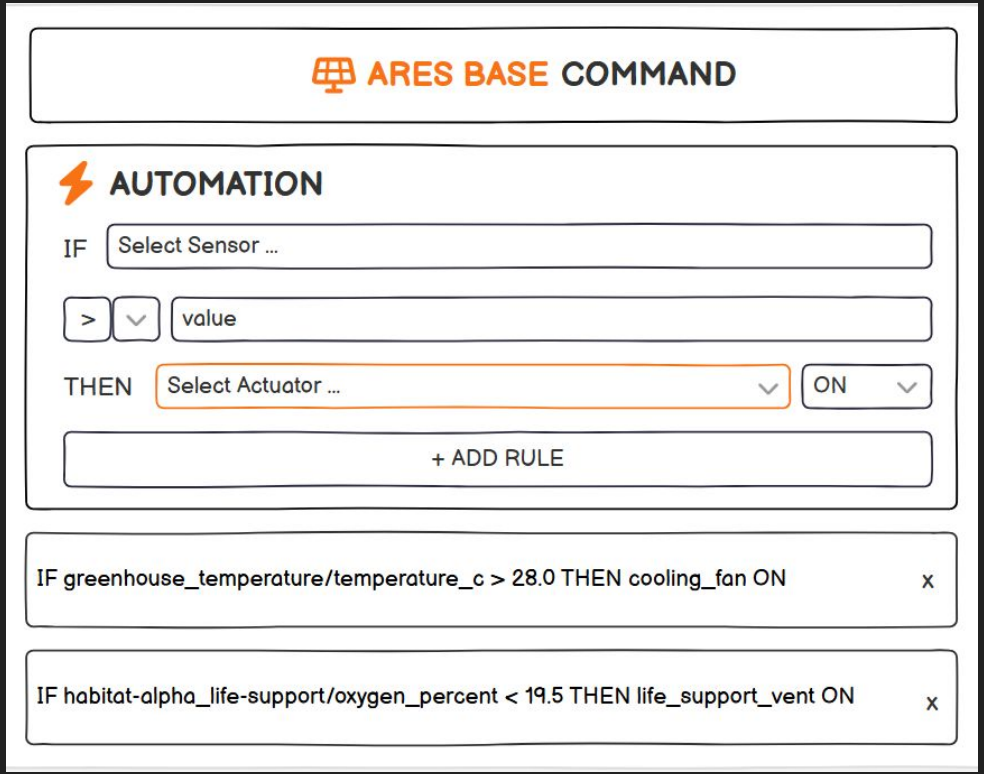


As a System Administrator, I want manual actuator toggles to trigger a REST POST request to the simulator so that the external physical system is updated correctly.

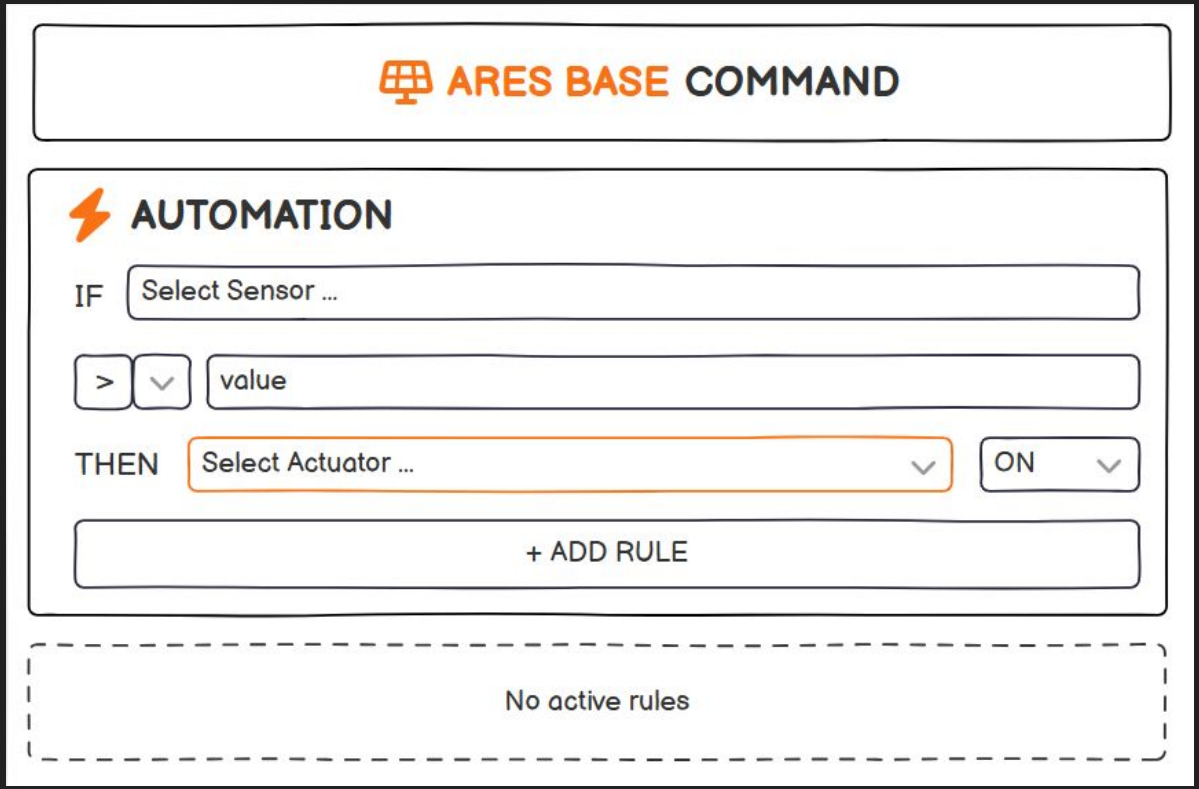
(NFR: The UI must update the actuator state via an asynchronous REST POST request, maintaining Command-Query separation)



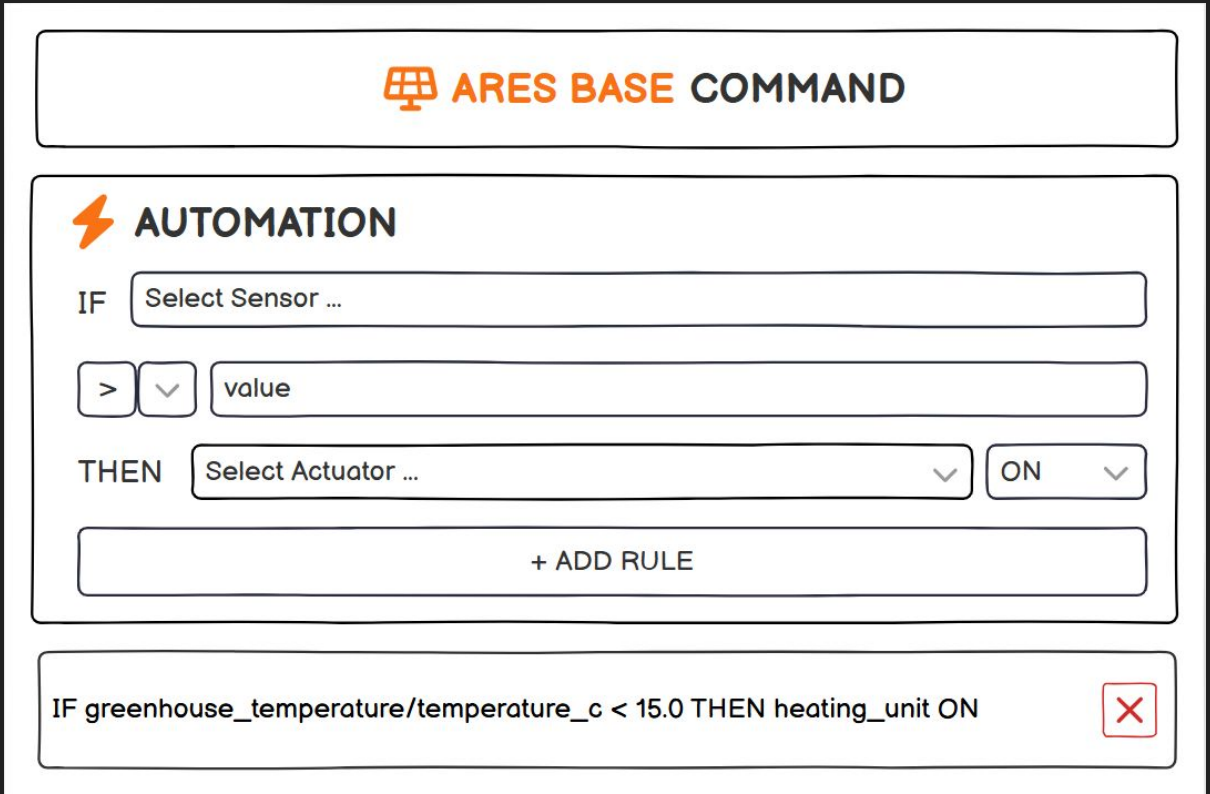
As a Habitat Operator, I want to view a list of all active automation rules via the frontend so that I know what automated controls are governing the base.



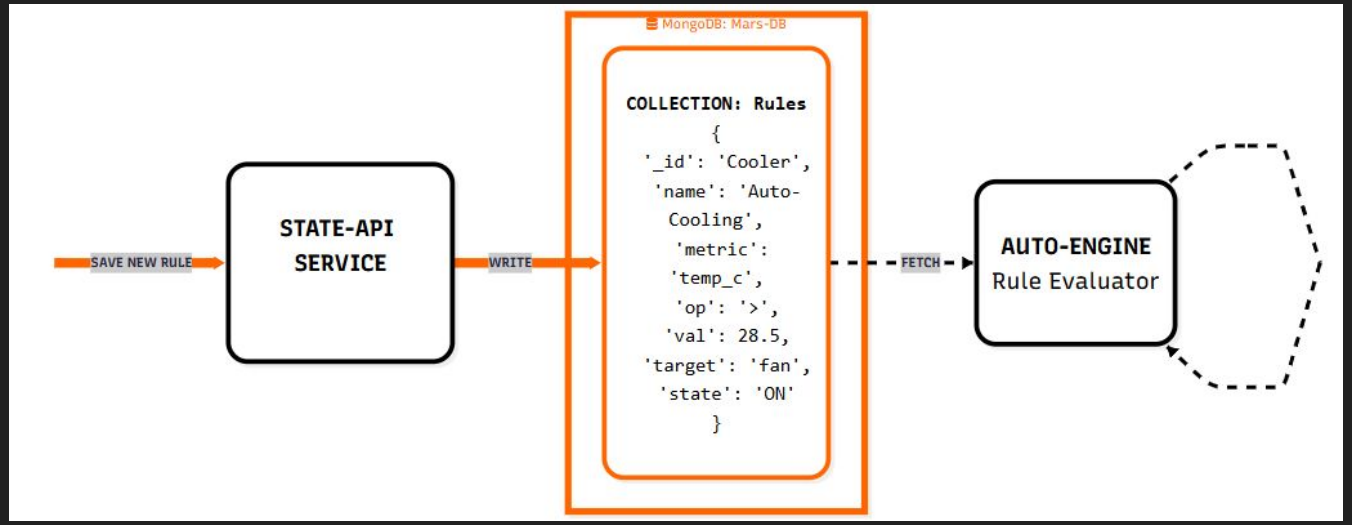
As a Habitat Operator, I want a frontend form to create a new IF-THEN rule so that the system can adapt to new environmental conditions.



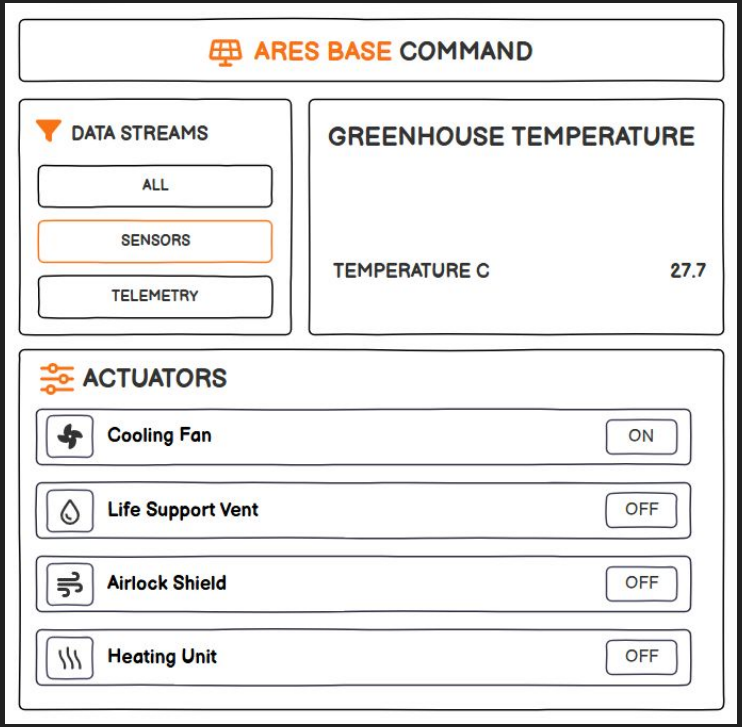
As a Habitat Operator, I want to be able to delete an existing rule from the interface so that I can remove obsolete automations.

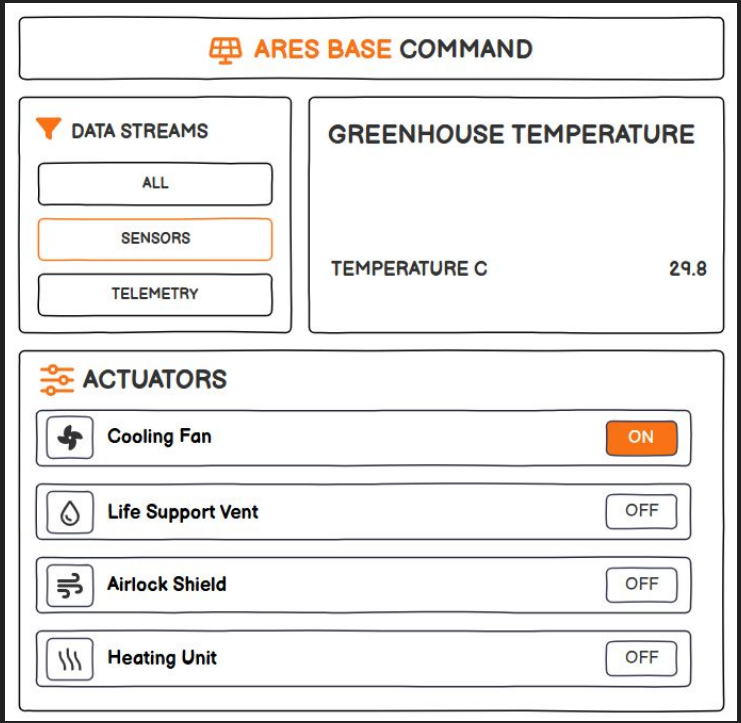


As a System Administrator, I want automation rules to be persisted in a database (MongoDB) so that they survive a system restart. **(NFR: Rules must be saved in a persistent database to survive container crashes or docker-compose down)**

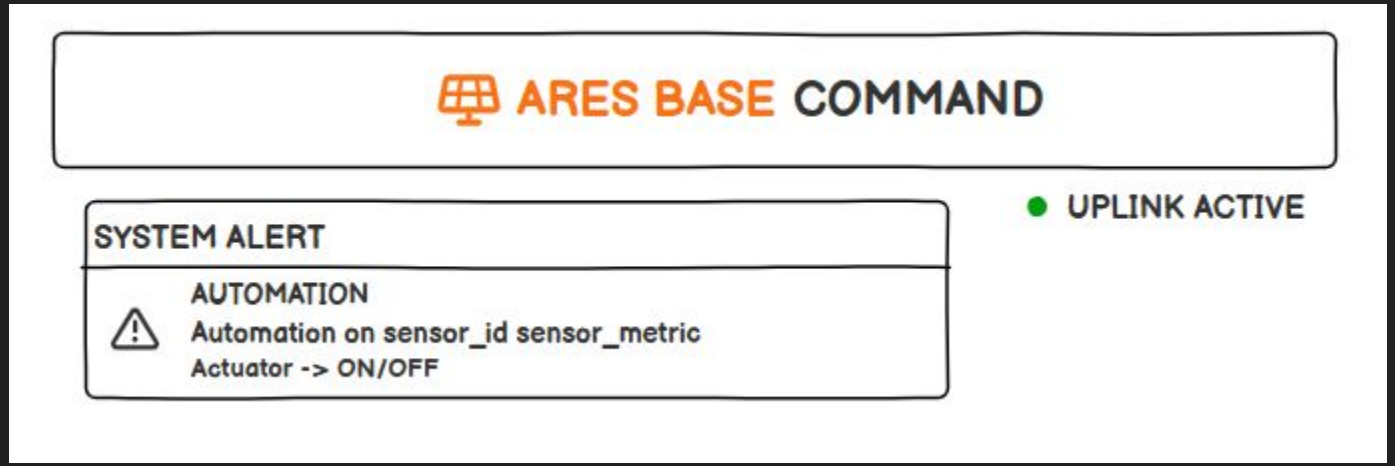


As a Habitat Operator, I want the dashboard to automatically update an actuator's UI toggle when it is modified by an automation so that I see the real-time status without refreshing. **(NFR: The UI must update via Socket.io without refreshing the browser)**

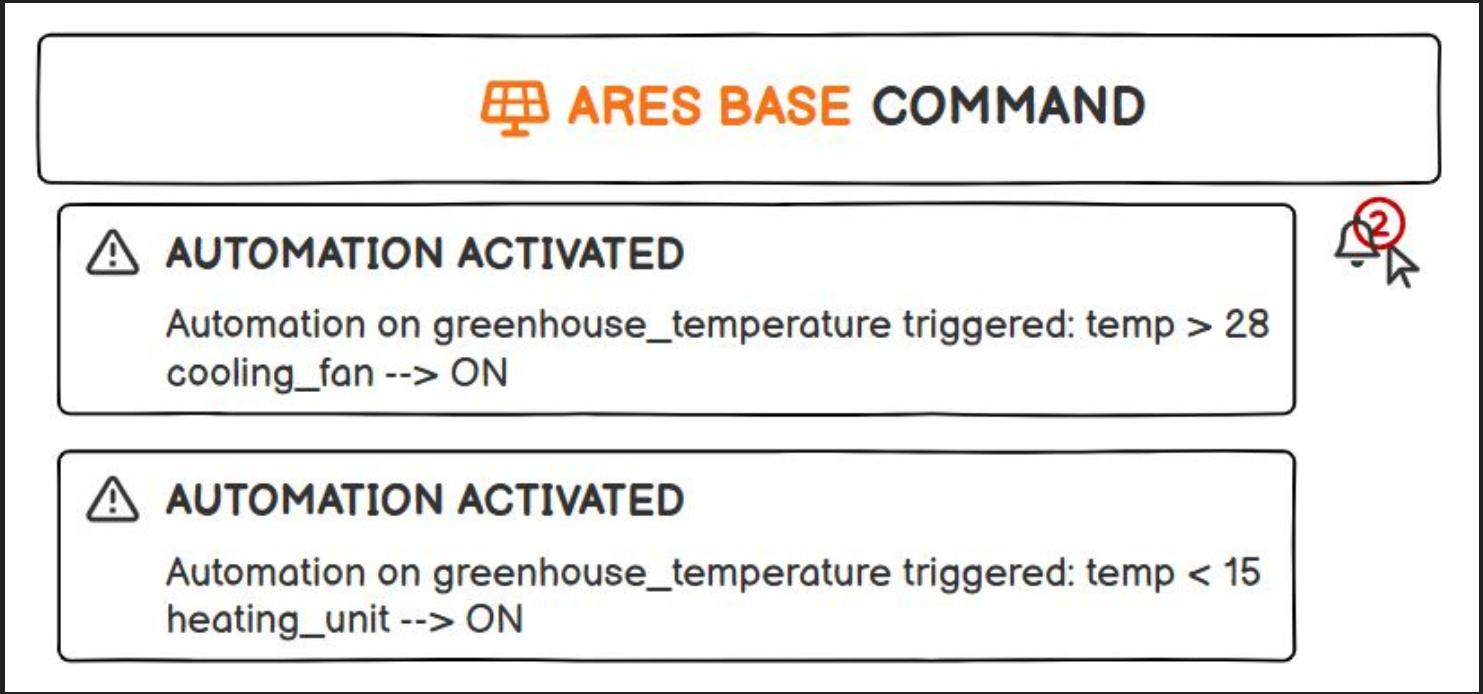




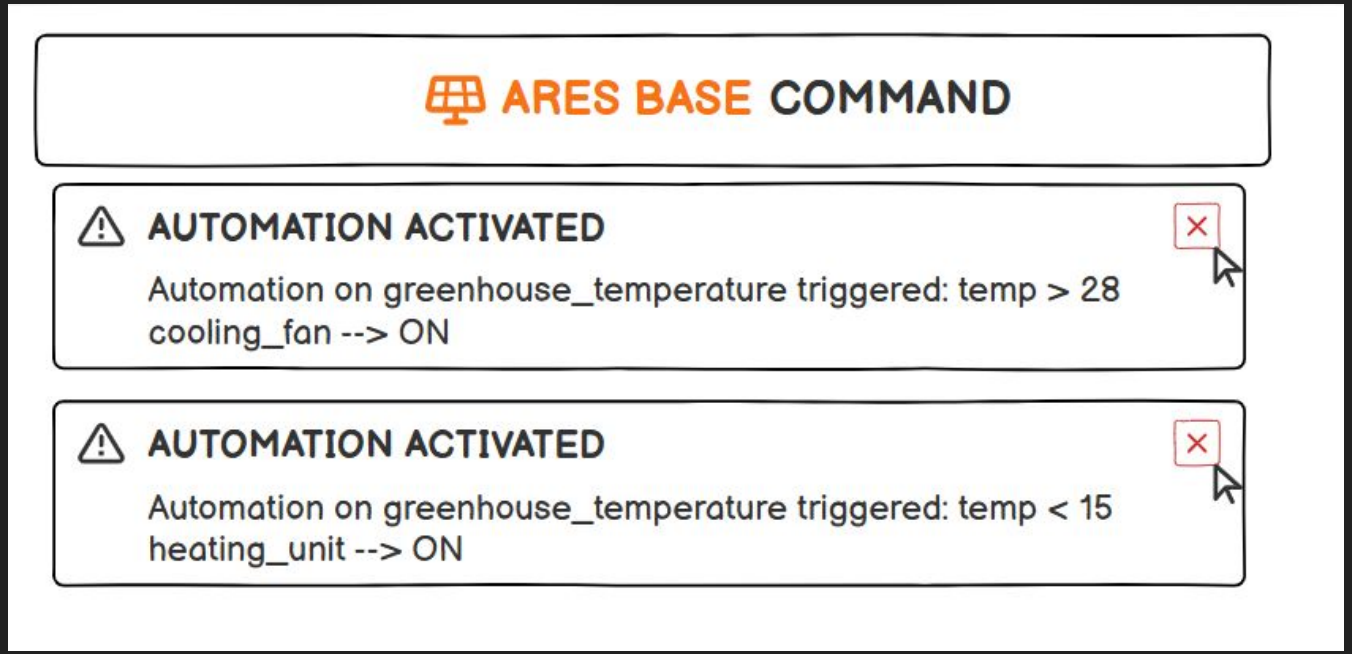
As a Habitat Operator, I want to receive real-time alert notifications on the dashboard when a rule triggers so that I am immediately notified of the autonomous action.



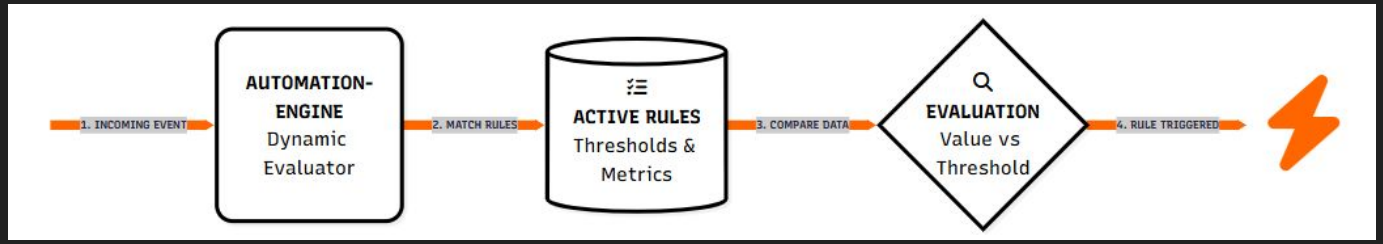
As a Habitat Operator, I want a dedicated notification dropdown menu on the dashboard so that I can review the history of recent alerts.



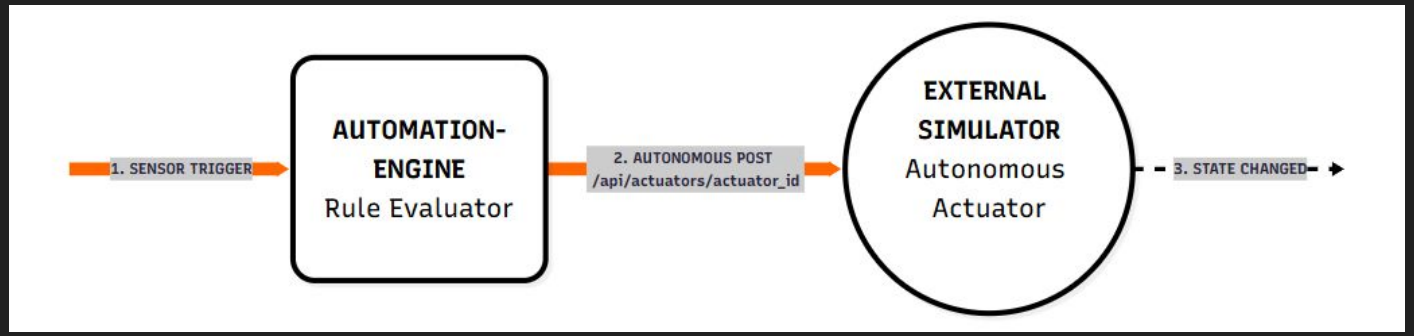
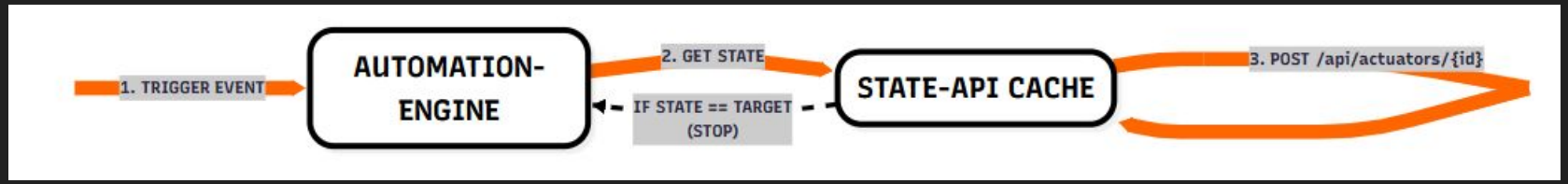
As a Habitat Operator, I want to be able to dismiss individual alerts from the notification panel so that I can keep the interface clean and focus on unresolved issues.



As a System Administrator, I want the Automation Engine to evaluate incoming events dynamically against active rules so that thresholds are monitored constantly.



As a System Administrator, I want the Engine to automatically send a POST command to the actuator when a rule triggers so that the habitat reacts autonomously.



As a System Administrator, I want the Engine to check the current state of an actuator before triggering it so that the system avoids spamming the simulator with redundant commands.